

Routing Protocols — Guided Notes

Unit 1.8.2 | CompTIA Network+ N10-009 Obj. 1.8 | N10-008 1.8

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Distance vector protocols share _____ with directly connected neighbors at regular intervals. Routers only know _____ and _____ to each destination — not the full topology. Link state protocols share _____ with all routers in the area, so every router builds _____.

Question 2

RIP uses _____ as its only metric. Its maximum is _____ hops — a destination _____ hops away is considered unreachable. RIP sends full table updates every _____ seconds, which causes slow _____.

Question 3

OSPF uses _____ as its metric, derived from _____. This means OSPF prefers _____ links over _____ links, unlike RIP which only counts _____.

Question 4

In OSPF, _____ is the backbone area that all other areas must connect to. Using areas limits the scope of _____ flooding and _____ calculations, allowing OSPF to scale to large networks.

Question 5

On an Ethernet segment with multiple OSPF routers, a _____ (DR) and _____ (BDR) are elected. All other routers form adjacencies only with these two. This reduces the number of _____ relationships from exponential to manageable.

Question 6

EIGRP's DUAL algorithm maintains a _____ — a pre-calculated backup route. If the primary route fails and one exists, EIGRP installs it _____ with no recalculation. EIGRP only sends updates when

_____, making it bandwidth-efficient.

Question 7

BGP is classified as a _____ protocol. It routes between _____ (ASes), each identified by an _____. _____ BGP runs between different ASes (e.g., your router to your ISP). _____ BGP runs within the same AS.

Question 8 — Real World Application

REAL WORLD: You inherit a network with three sites running RIP. Users at Site C (which is 14 hops from Site A) report they can reach Site A fine, but a new Site D has been added 2 hops beyond Site C. Users at Site D cannot reach Site A at all. Explain why and what you would do.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-8-2-routing-protocols/1-8-2-routing-protocols-answer-key.pdf
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